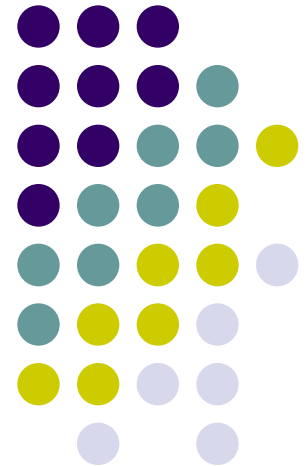
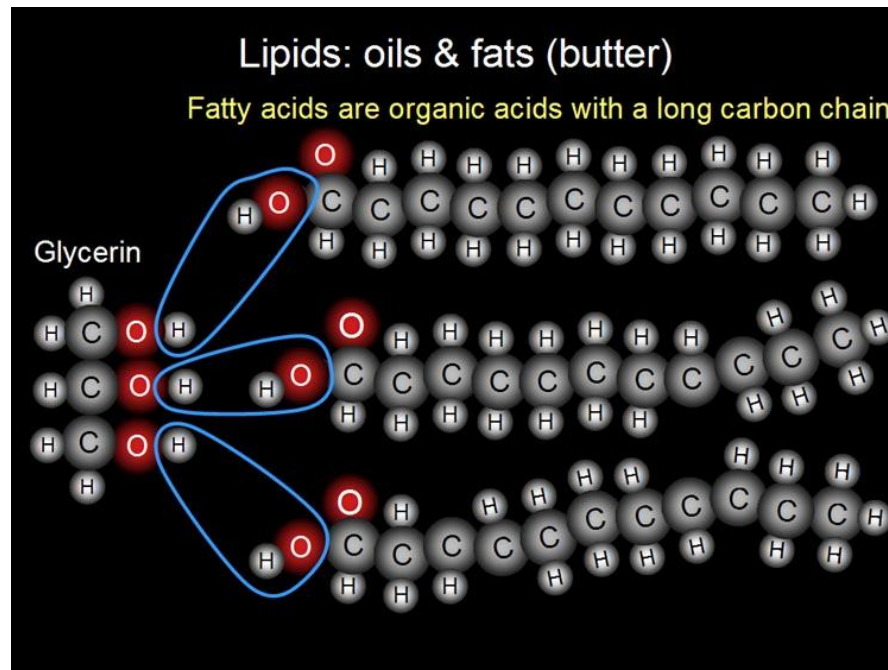
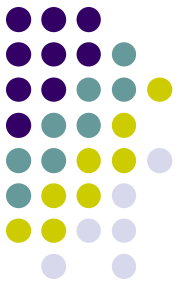


LIPIDS

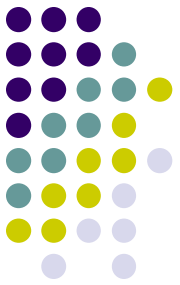


OBJECTIVES

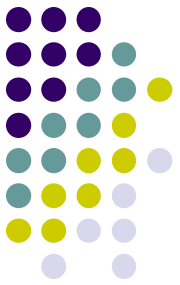


- At the end of the lecture student would be able to :
 - a. Define lipids.
 - b. Describe classification of fatty acids.
 - c. Understand the biological significance of lipids.

WHAT ARE LIPIDS?

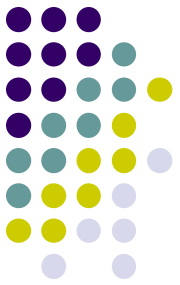


- ❑ The lipids constitute a very important heterogeneous group of organic substances, including fats, oils, steroids, waxes, and related compounds.
- ❑ Related to fatty acids.
- ❑ Chemically they are various types of esters of different alcohols.
- ❑ Formed of long-chain hydrocarbon groups, also contain oxygen, phosphorus, nitrogen and sulfur.
- ❑ Utilized by living organisms.



BLOOR'S CRITERIA

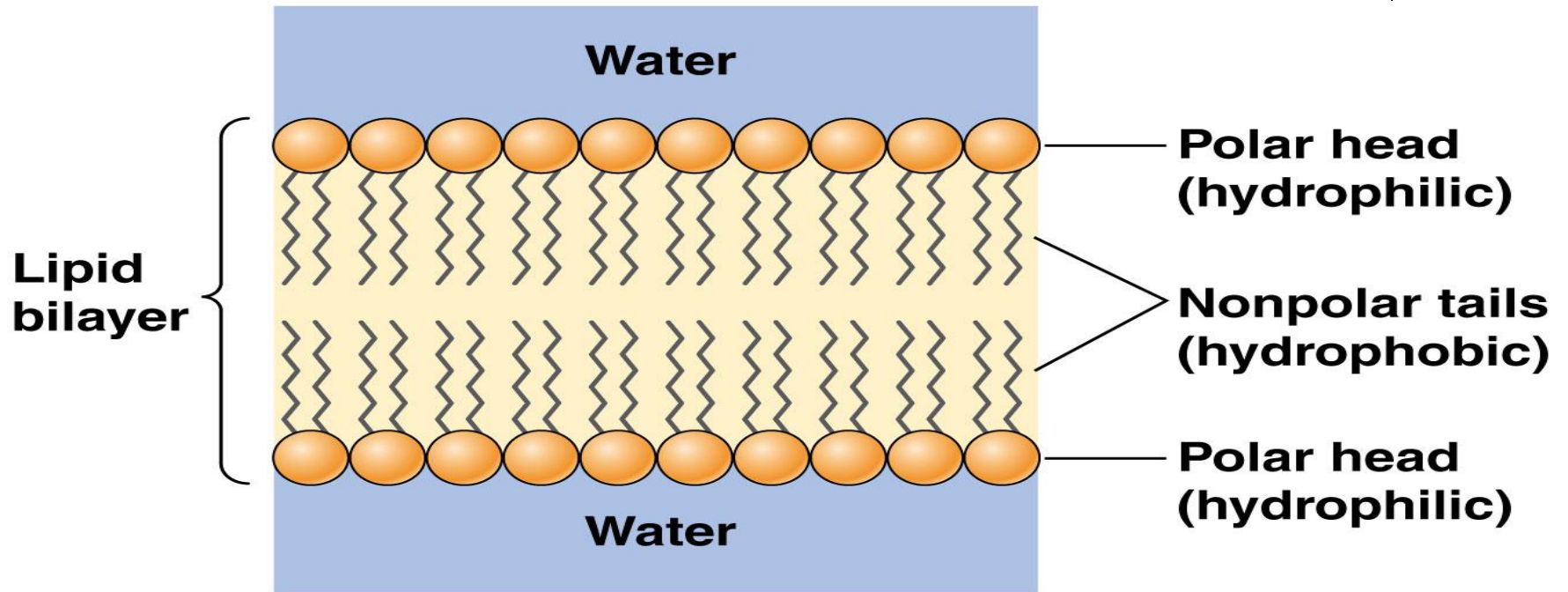
- According to Bloor:
 - They are “**INSOLUBLE IN WATER**”
“**SOLUBLE IN FAT SOLVENTS**”,
Such as ether, chloroform,
benzene, acetone, etc.



Lipids are non-polar (hydrophobic) compounds

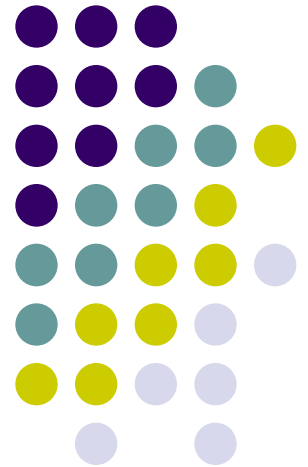
BUT

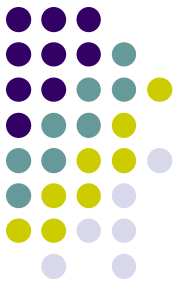
Lipid molecules in cell membranes are amphipathic (or amphiphilic)—that is, they have a hydrophilic (“water-loving”) or *polar* end and a hydrophobic (“water-fearing”) or *nonpolar* end.



Functions of Lipids

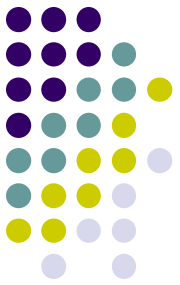
or
Biomedical Importance
of Lipids



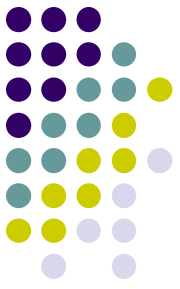


The functions of lipids are given below:

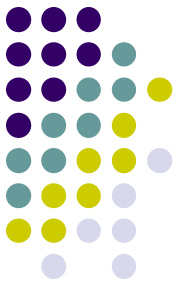
- They are good sources of energy i.e 9.1 kilocalories per gram fat utilized in the body.
- Lipids in food also act as carriers of fat soluble vitamins and essential fatty acids.
- They also make food more palatable and serve to decrease its mass.
- The dietary lipids decrease gastric motility and secretions and have a high satiety value.
- Body fat provides contour to the body and also gives anatomical stability to organs like kidneys .



- When a person loses weight rapidly ,his kidneys are liable to become floating kidneys.
- Fats are good energy reservoirs in the body. Adipose tissue is best suited for this purpose.
- Lipids exert an insulating effect on the nervous tissue.
- Lipids are an integral part of cell membranes.
- Some lipids act as precursors of very important physiological compounds . For example , cholesterol is the precursor of steroid hormones while 7-dehydrocholesterol is the precursor of vitamin D3.

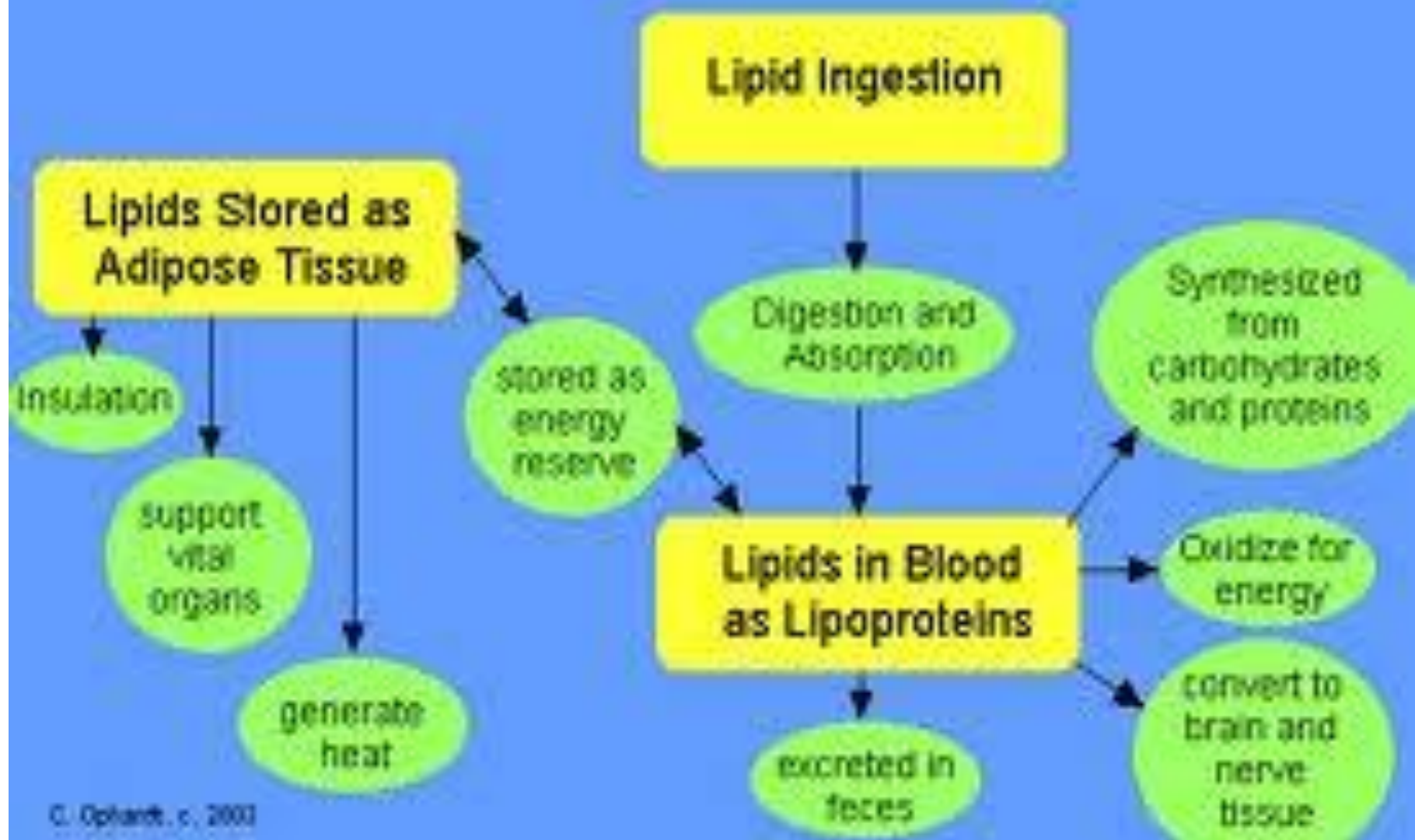


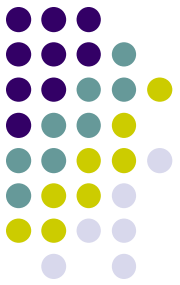
- Also these lipids help to prevent water evaporation from the skin ; with out this protection the amount of the daily insensible perspiration would probably be 15 to 20 liters instead of the usual 300 to 400 ml.
- The sphingosine – containing lipid have a role in the transmission of the nerve impulses across synapses.
- Prostaglandins and their related compounds have hormone like actions in the body and very important physiological compounds.
- In certain cases derivatives of lipids act as intracellular messengers after they are released .



- Concerned with cell recognition, species specificity and tissue immunity.
- Bile salts are derived lipids that help the digestion and absorption of lipids.
- Lipoproteins (complexes of lipids and proteins) helps in the transport of lipids in plasma.

Lipid Function and Metabolism Summary



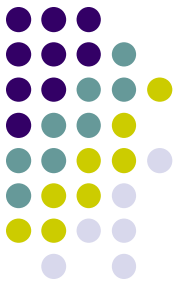


Clinical significance of lipids

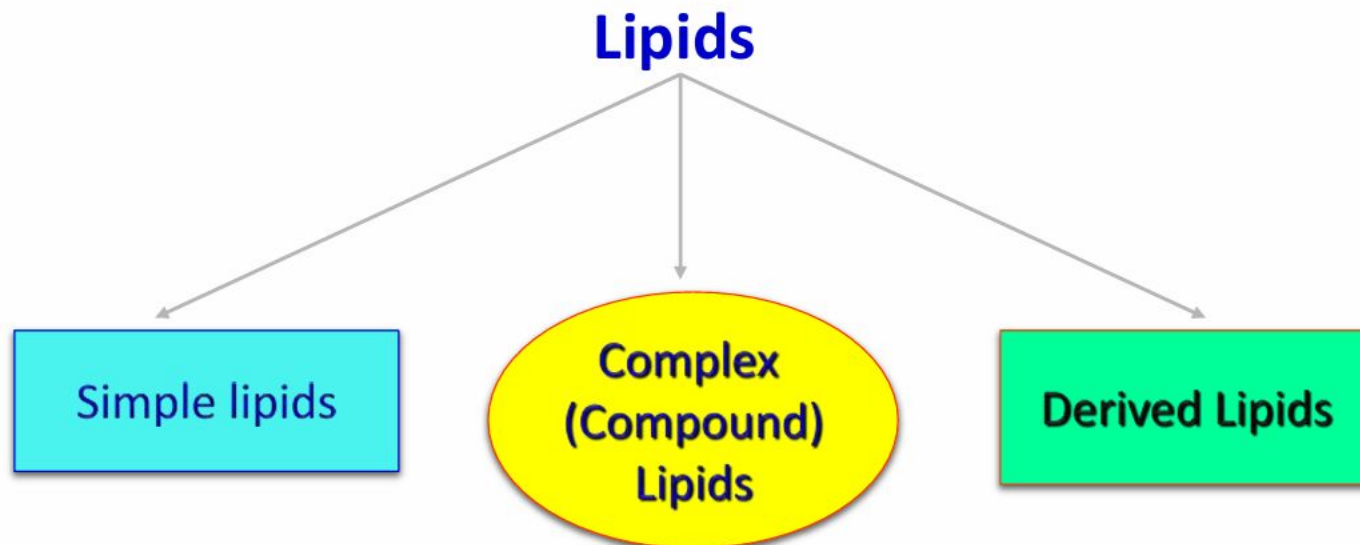
Following diseases are associated with abnormal chemistry or metabolism of lipids-

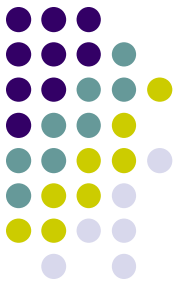
- ❑ Obesity**
- ❑ Atherosclerosis**
- ❑ Diabetes Mellitus**
- ❑ Hyperlipoproteinemia**
- ❑ Fatty liver**
- ❑ Lipid storage diseases**

CLASSIFICATION OF LIPIDS



- They are broadly classified based on their chemical composition into :
 1. Simple lipids
 2. Compound lipids
 3. Derived or associated lipids

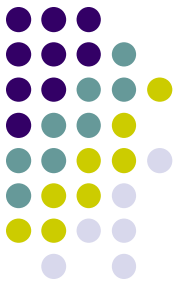




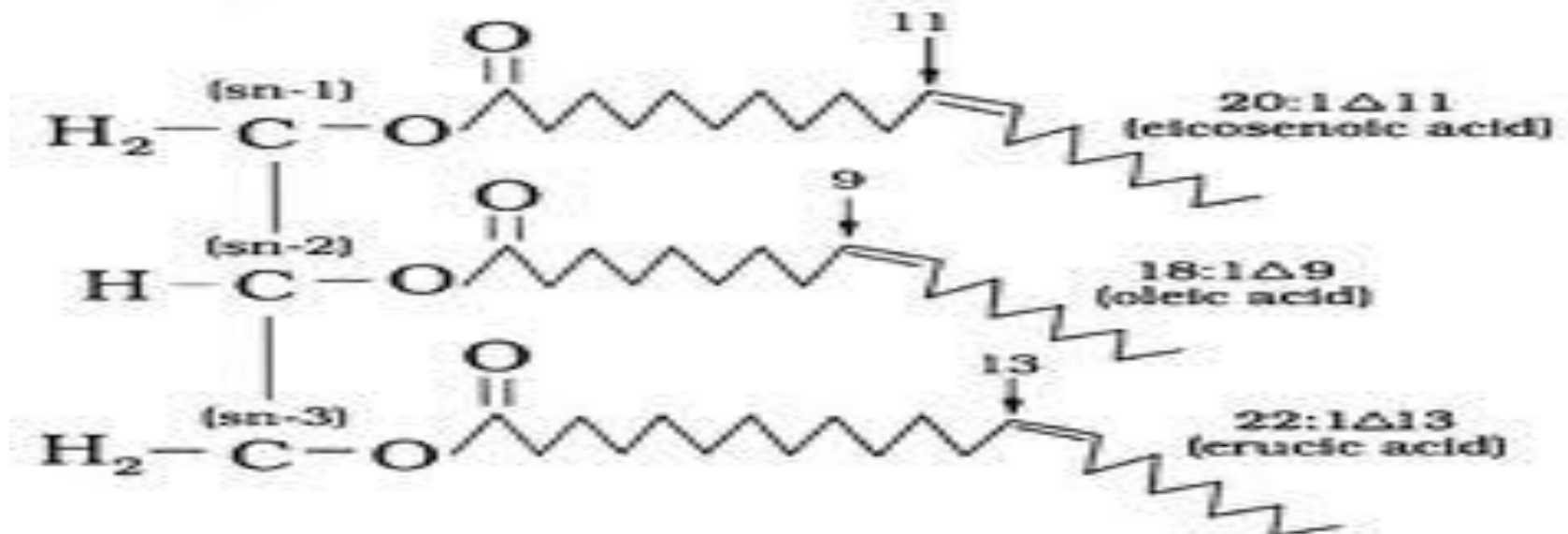
SIMPLE LIPIDS

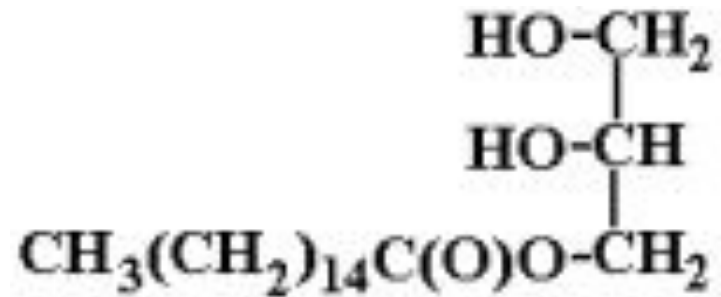
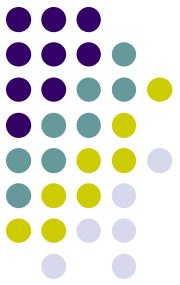
- Simple lipids contain C, H, and O.
- These consist of following subgroups:
- **1:FATS :**
- Synonyms: Neutral fat , Triglyceride or Triacylglycerol.
- Def: These are esters of fatty acids with glycerol.

- **Fats** and **oils** are made from two kinds of molecules: **glycerol** (a type of alcohol with a hydroxyl group on each of its three carbons) and three **fatty acids**. Since there are three fatty acids attached, these are known as **triglycerides**.
- The main distinction between fats and oils is whether they're solid or liquid at room temperature, and this, is based on differences in the structures of the fatty acids they contain.

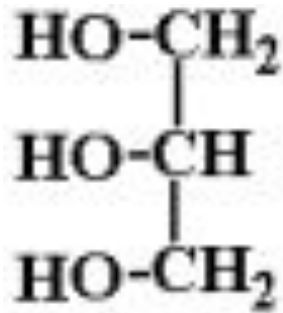


TRIACYLGLYCEROL

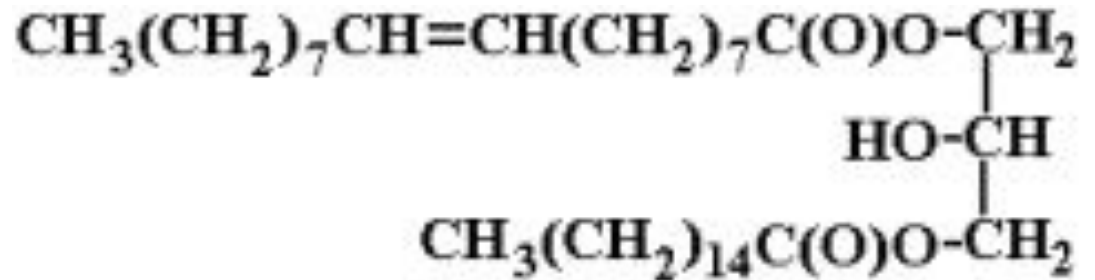




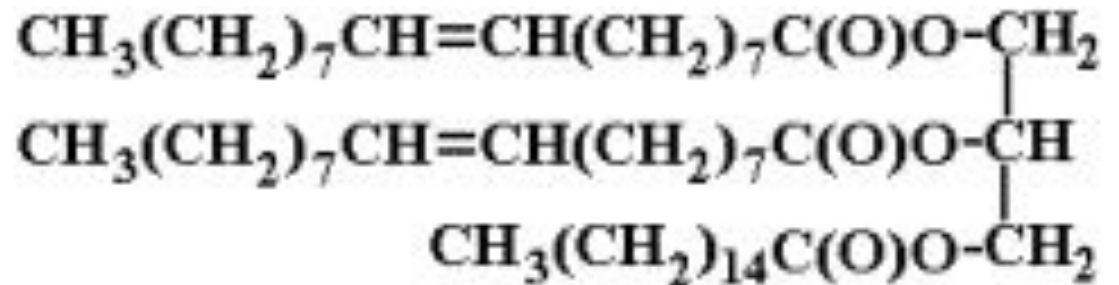
Monoglyceride



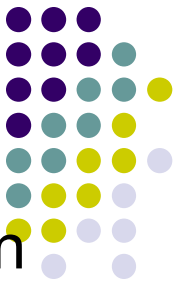
Glycerol



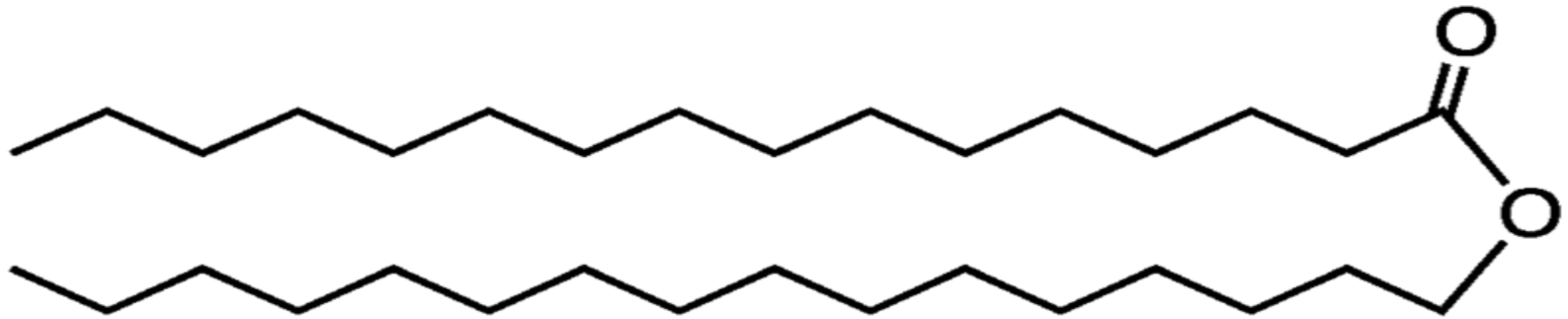
1,3 Diglyceride may be 1,2



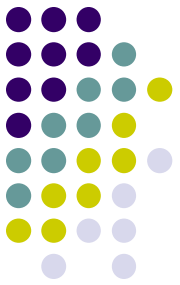
Triglyceride



- **2:WAXES:**
- These are esters of fatty acids with alcohols other than glycerol (monohydric long chain alcohols (C16 - 30)).
- Eg. Bees wax, lanolin, spermaceti oil.
- Used for lotions, ointments.

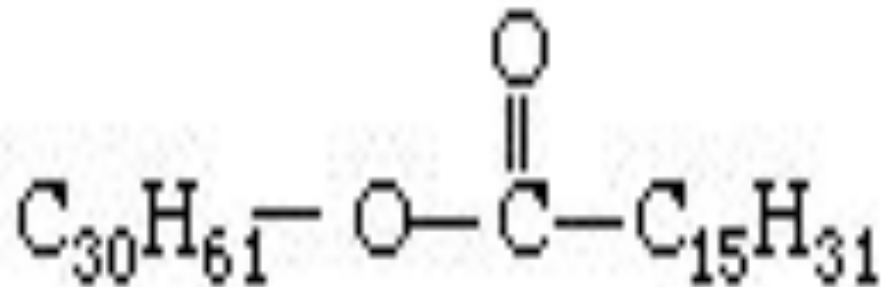


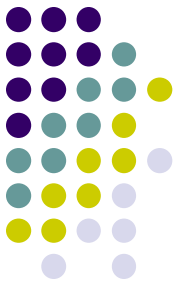
Wax (Cetyl palmitate)



- **Animal waxes:**
- **Beeswax** is a natural wax produced in the bee hive of honey bees. It is mainly esters of fatty acids and various long chain alcohols.

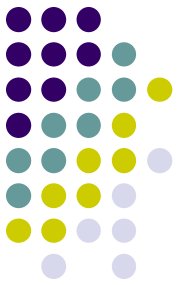
Beeswax (myricyl palmitate)





COMPOUND LIPIDS

- Compound lipids contain one or more additional elements, such as phosphorus, nitrogen, or sulfur along with carbon hydrogen and oxygen.
- “Esters of fatty acids with alcohol + containing additional groups (prosthetic group)”.
- Include in it
 - I. Glycerophospholipid
 - II. Sphingolipids
 - III. Lipoproteins
 - IV. Lipopolysaccharides

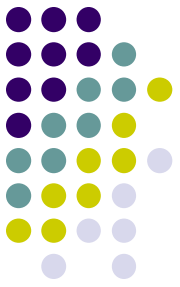


DERIVED LIPIDS

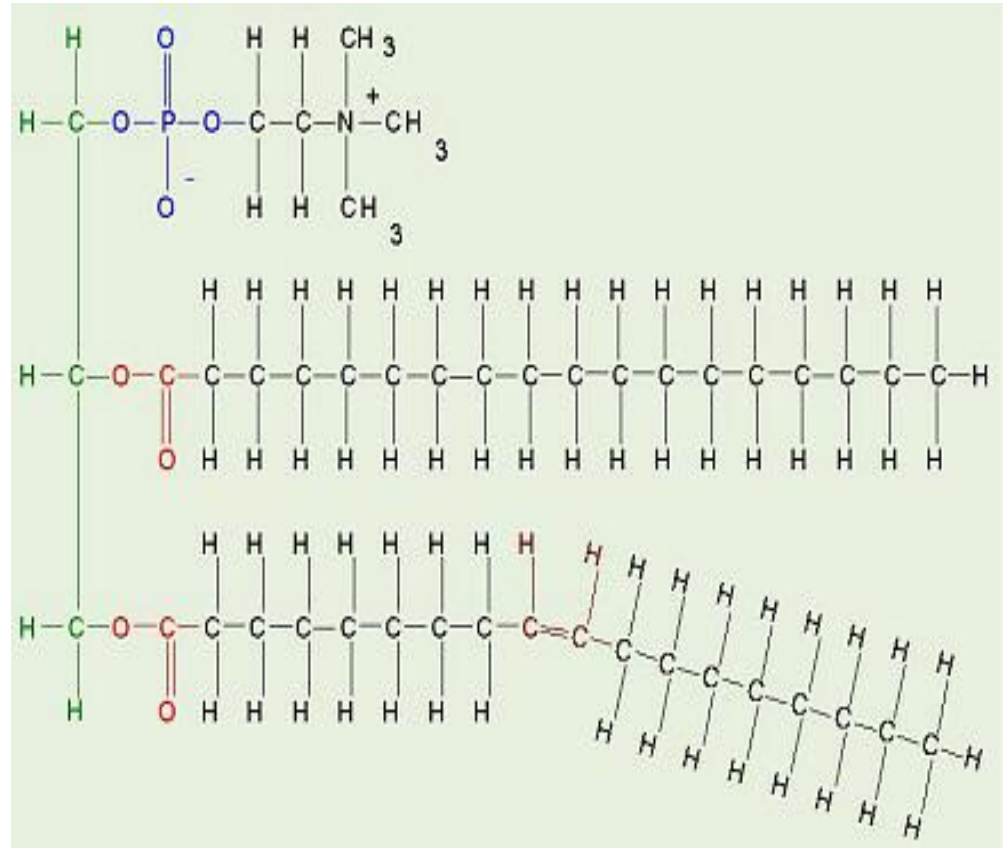
Derived from simple lipids or complex lipids on hydrolysis.

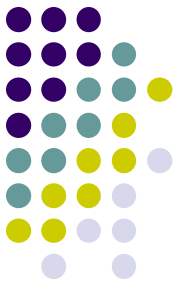
- Fatty acids, glycerol.
- Monoacyl and Diacylglycerol.
- Steroids- lipids with cyclopentanoperhydrophenanthrene (steroid) ring.
 - E.g. – cholesterol and its derivatives- bile acids, hormones, Ergosterol.
- Eicosanoids – Arachidonic acid derivatives- PG, TXA₂,

A:Glycerophospholipids

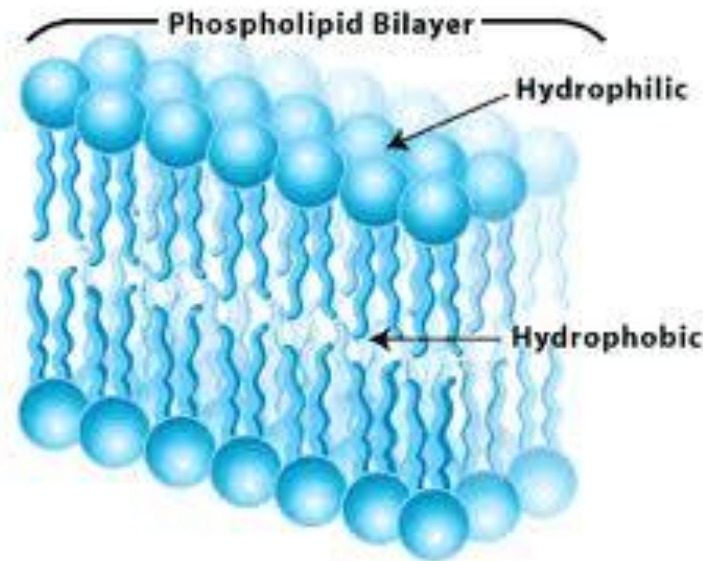


GLYCERO PHOSPHOLIPIDS are made from glycerol, two fatty acids, and (in place of the third fatty acid) a **phosphate group** with some other molecule attached to its other end. The hydrocarbon tails of the fatty acids are still hydrophobic, but the phosphate group end of the molecule is hydrophilic because of the oxygens with all of their pairs of unshared electrons. This means that phospholipids are soluble in both water and oil (amphiphilic or amphipathic).



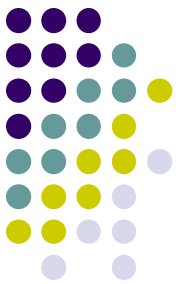
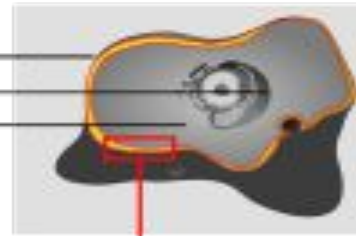


They are the main component of biological membranes. Our **cell membranes** are made mostly of phospholipids arranged in a double layer with the tails from both layers “inside” (facing toward each other) and the heads facing “out” (toward the watery environment) on both surfaces.

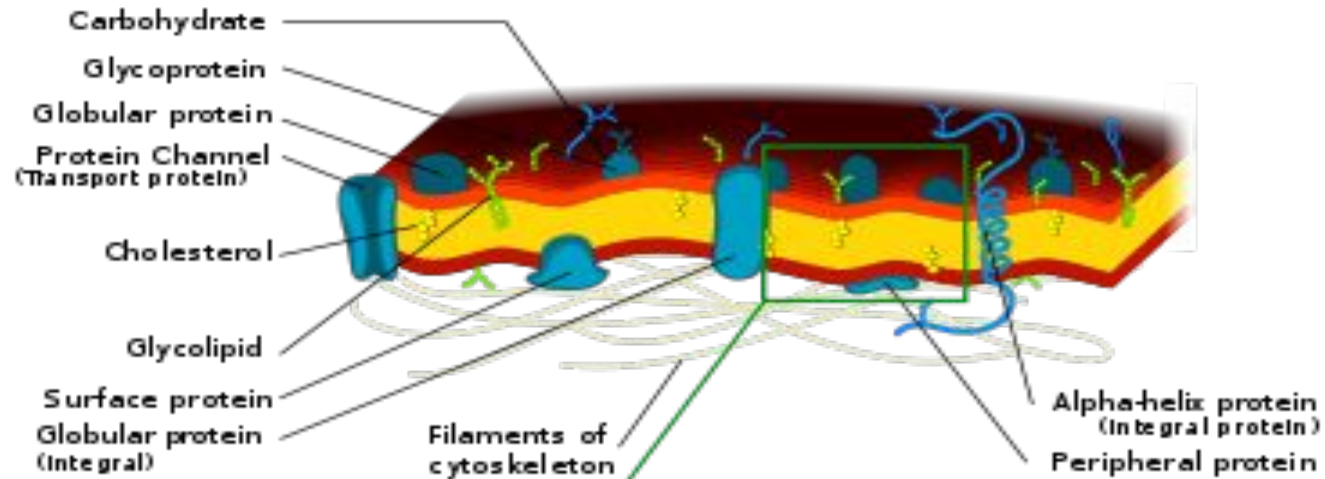


Cell

Extracellular fluid
Nucleus
Cytoplasm



Cell membrane



Phospholipid bilayer

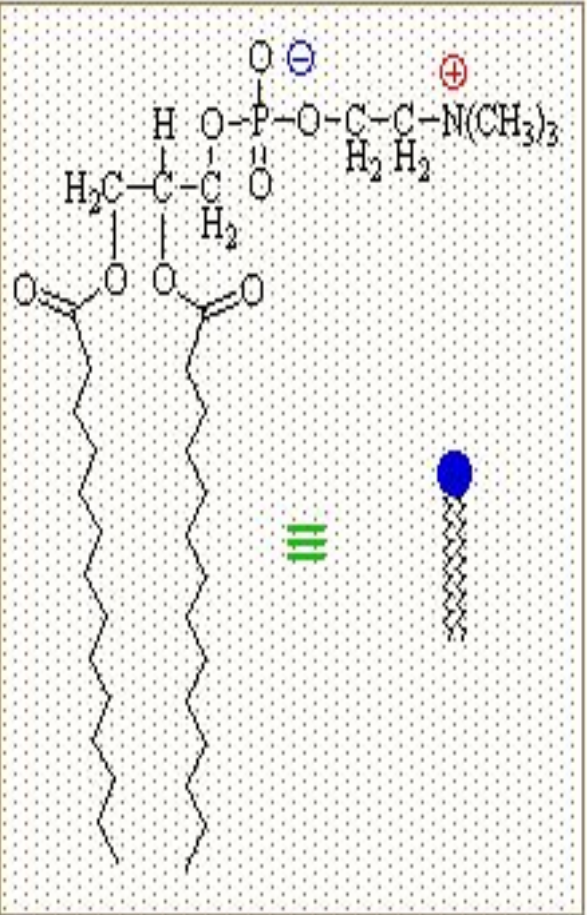


Phospholipid (Phosphatidylcholine)



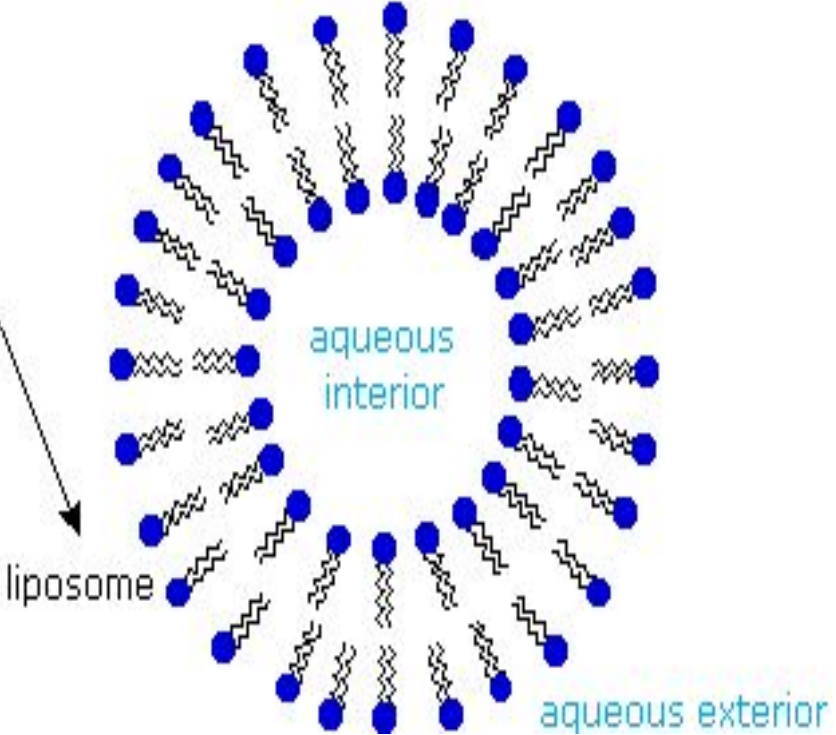
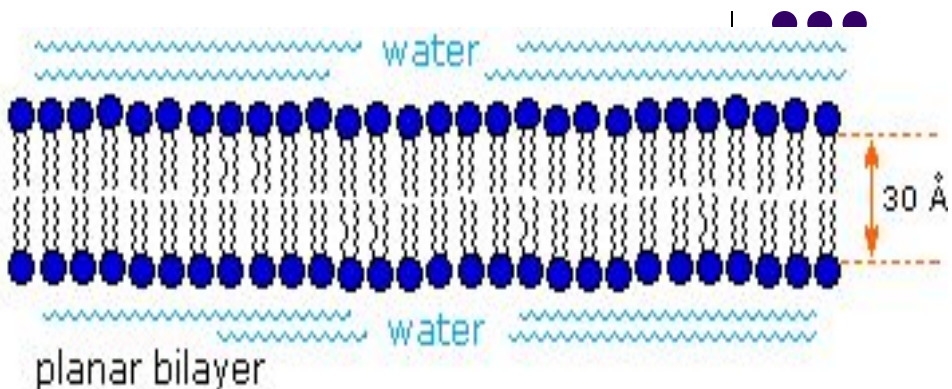
Hydrophilic head

Hydrophobic tail

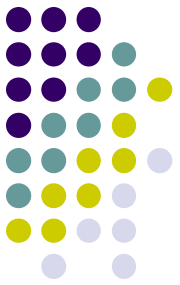


phospholipid

aggregation
in water

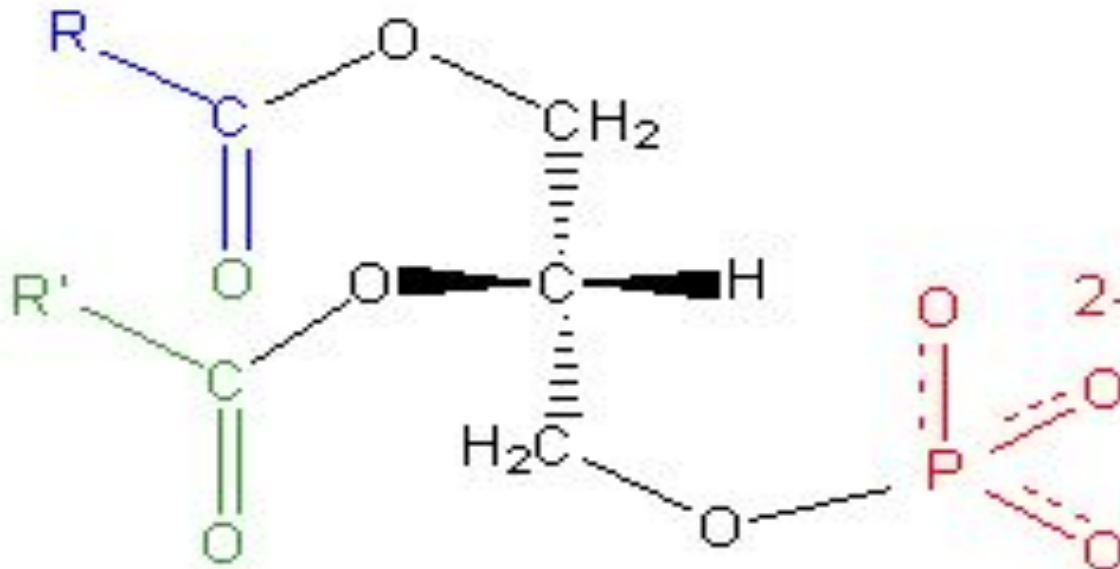


Types of Phosphoglycerides or Glycerophospholipids



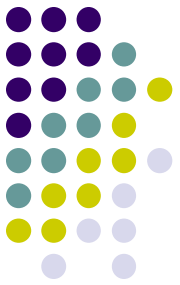
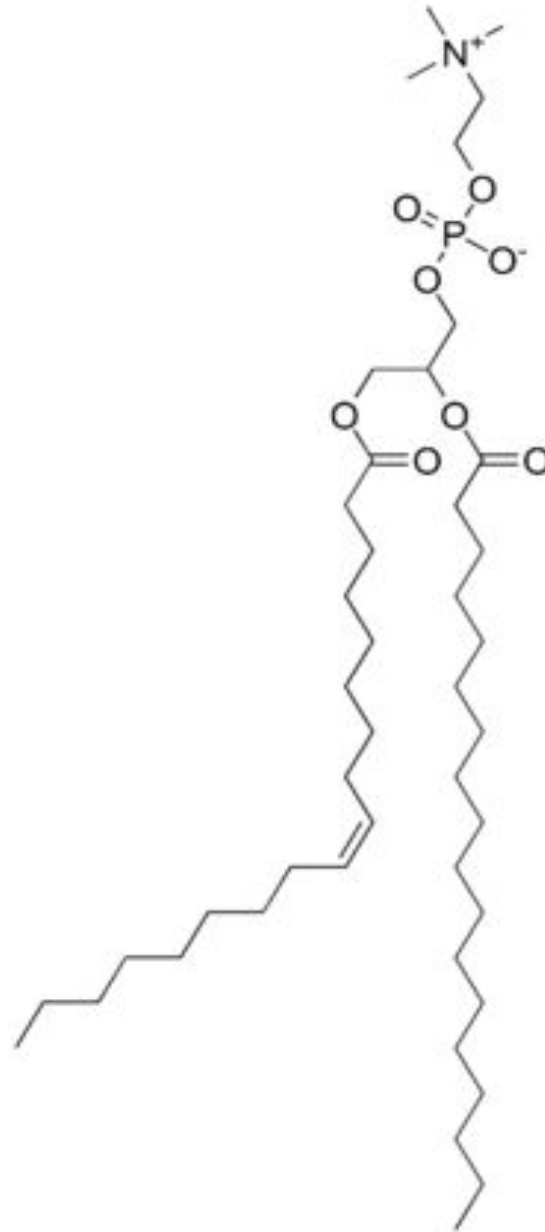
- **1:PHOSPHATIDIC ACID:**

- Phosphatidic acid consists of a glycerol backbone, with, in general, a fatty acid bonded to carbon-1, an fatty acid bonded to carbon-2, and a phosphate group bonded to carbon-3.

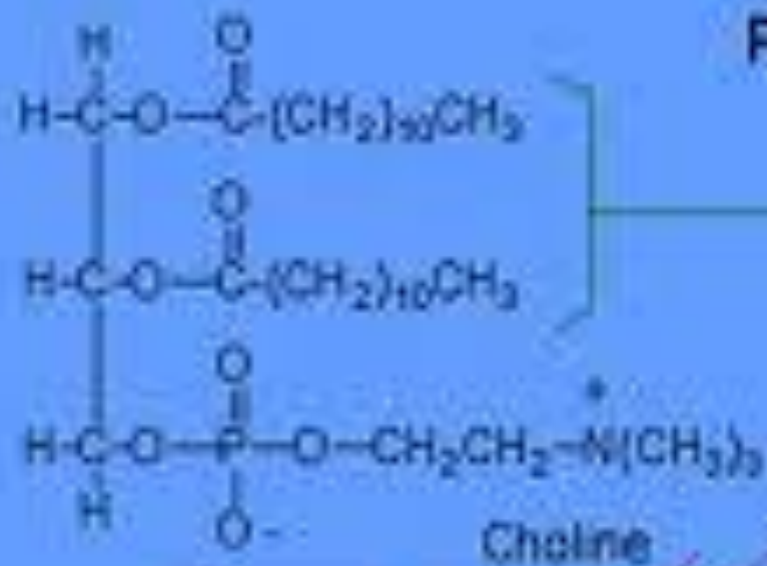


2.LECITHINS

- **Lecithin**(phosphatidylcholine) is yellow-brownish fatty substances occurring in animal and plant tissues, and in egg yolk, composed of phosphoric acid, choline(nitrogenous base), fatty acids, glycerol .



Lecithin Phospholipid



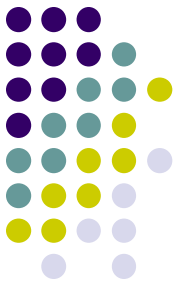
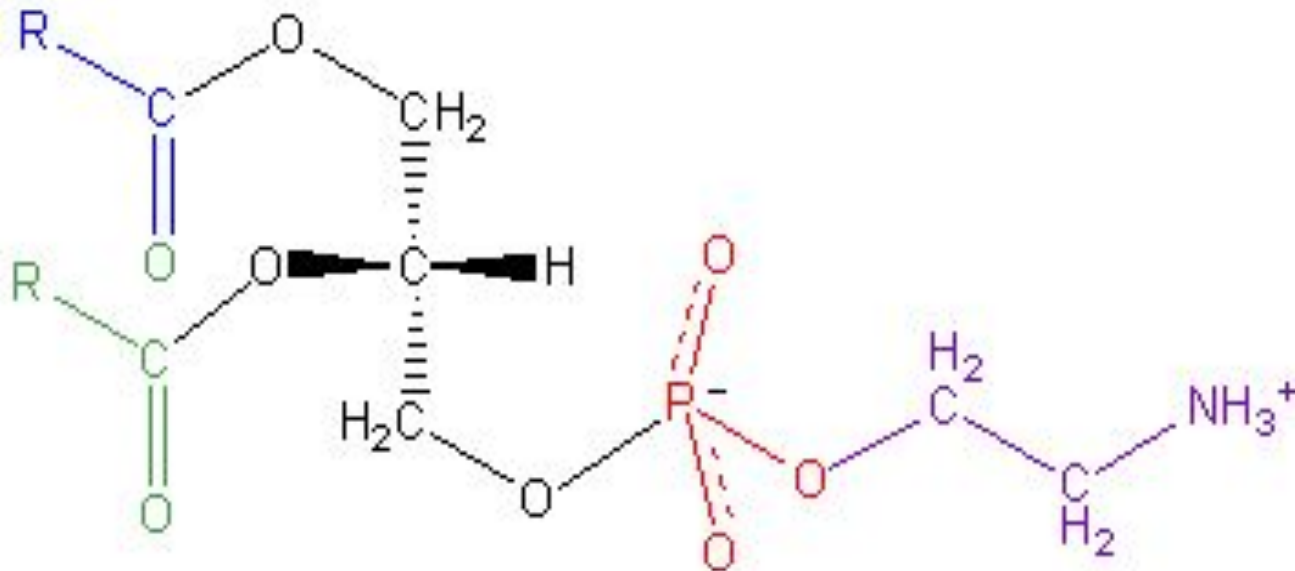
Non-Polar

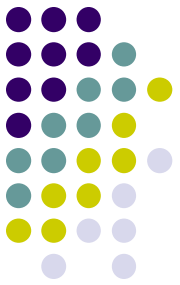
Polar



• 3:CEPHALINS:

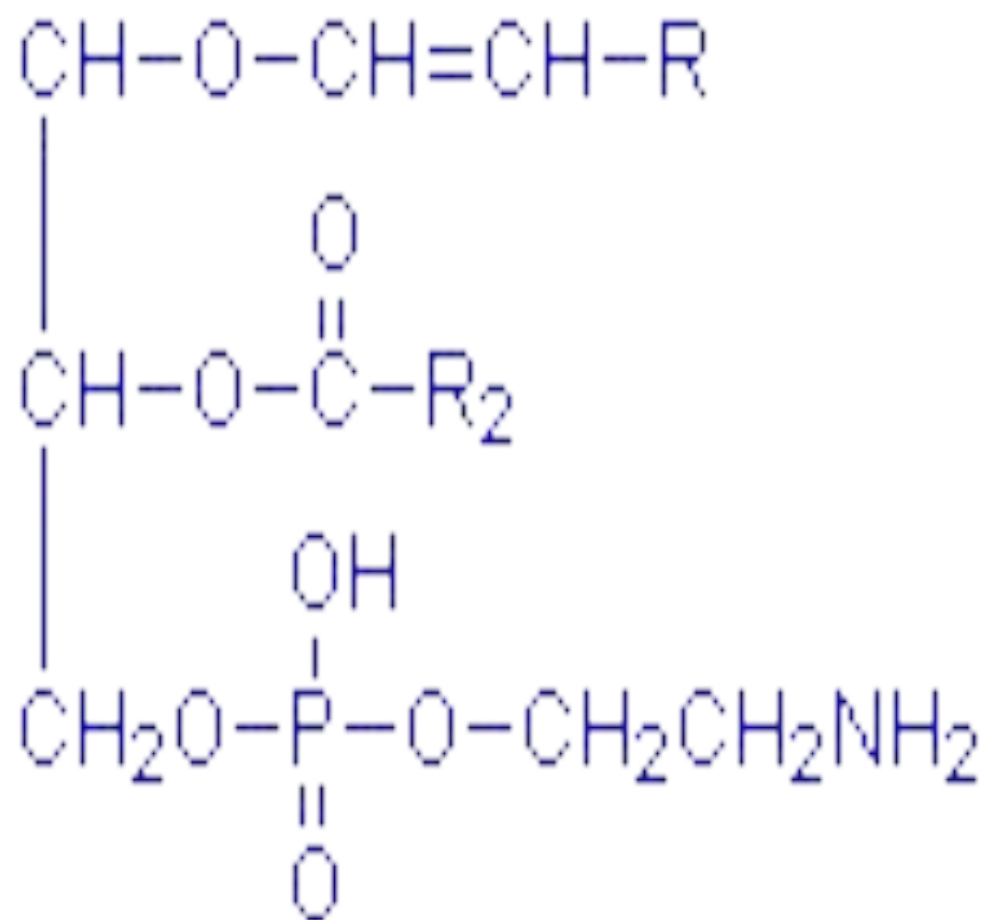
- **Cephalin** or **Phosphatidylethanolamine (PE)** is a phospholipid, found in biological membranes. It is synthesized by the addition of ethanolamine to diglyceride.
- Cephalin is found in all living cells, although in human physiology it is found particularly in nervous tissue such as the white matter of brain, nerves, neural tissue, and in spinal cord.

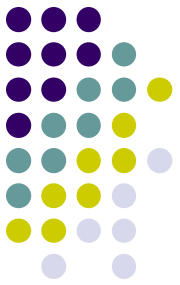




4: PLASMALOGEN

- A **plasmalogen** is an ether lipid where the first position of glycerol binds a vinyl residue (from a vinyl alcohol) with the double bond next to the ether bond.
- The second carbon has a typical ester-linked fatty acid, and the third carbon usually has a phospholipid head group like choline or ethanolamine.
- In many tissues plasmalogens are minor constituents, but in heart tissue nearly 50% of phosphatidylcholine contains the alkenyl ether at carbon 1. Nervous tissues, testes and kidneys also contain significant amounts of plasmalogens.





B:SPHINGOLIPIDS

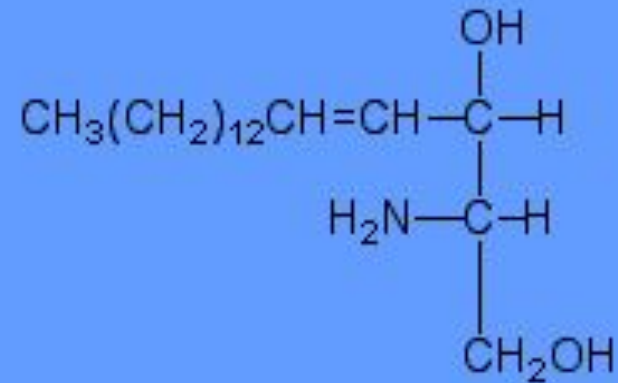
Sphingolipids are a second type of lipid found in cell membranes, particularly nerve cells and brain tissues. They do not contain glycerol, but retain the two alcohols with the middle position occupied by an amine called sphingosine.

Sphingosine and fatty acid combination is called CERAMIDE.



As shown in the graphic on the left, sphingosine has three parts, a three carbon chain with two alcohols and amine attached and a long hydrocarbon chain.

Sphingosine

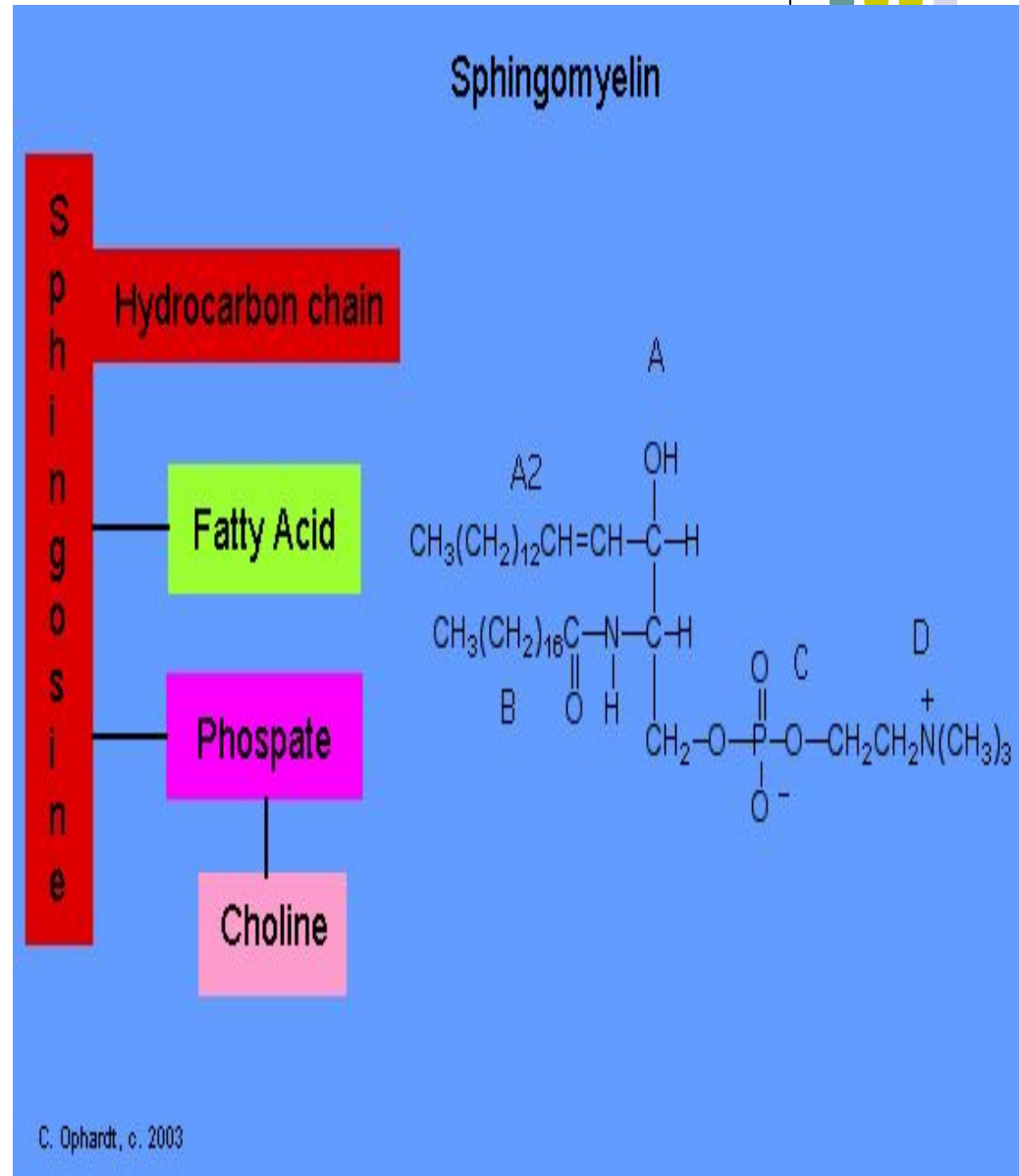


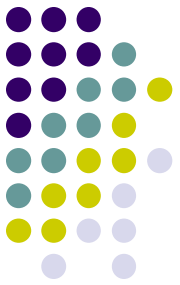
TYPES OF SPHINGOLIPIDS



1:SPHINGOMYELINS

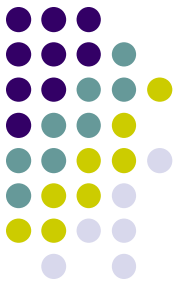
In sphingomyelin, the base sphingosine has several other groups attached. A fatty acid is attached to the amine through amide bond. Phosphate is attached through a phosphate ester bond, and again through a phosphate ester bond to choline.





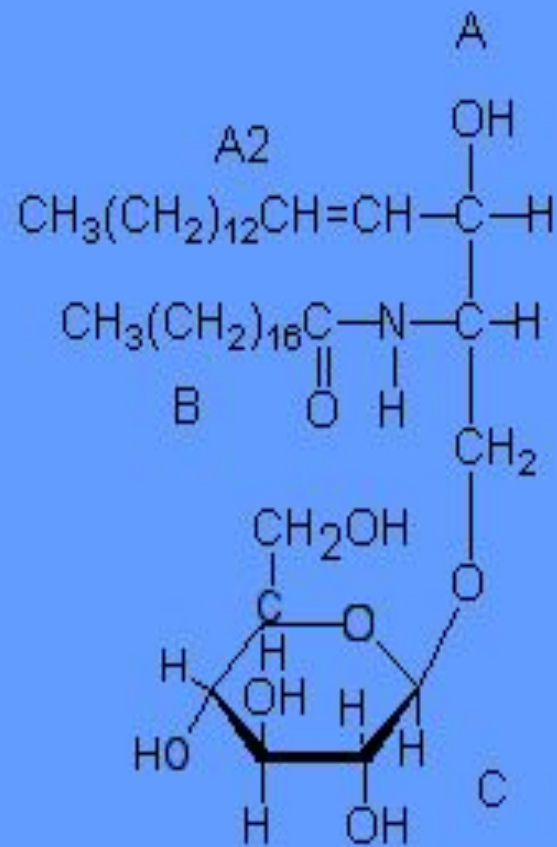
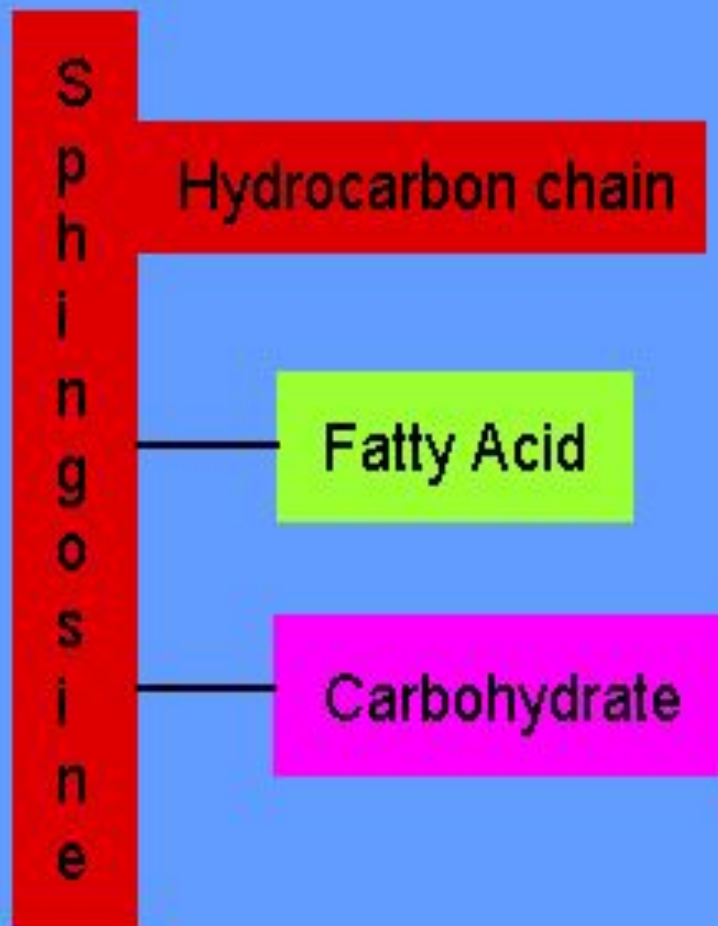
- The human brain and spinal cord is made up of gray and white regions. The white region is made of nerve axons wrapped in a white lipid coating, the myelin sheath, which provides insulation to allow rapid conduction of electrical signals.
- Sphingomyelins are located throughout the body in nerve cell membranes. They make up about 25 % of the lipids in the myelin sheath that surrounds and insulates cells of the central nervous system.

2:GLYCOLIPIDS

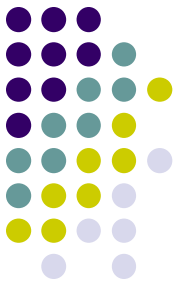


- Glycolipids are complex lipids that contain carbohydrates. Cerebrosides are an example which contain the sphingosine backbone attached to a fatty acid and a carbohydrate. The carbohydrates are most often glucose or galactose. Glucocerebroside has the specific function to be in the cell membranes of macrophages, (cells that protect the body by destroying foreign microorganisms. Galactocerebroside is found almost exclusively in the membranes of brain cells.

Glucocerebroside

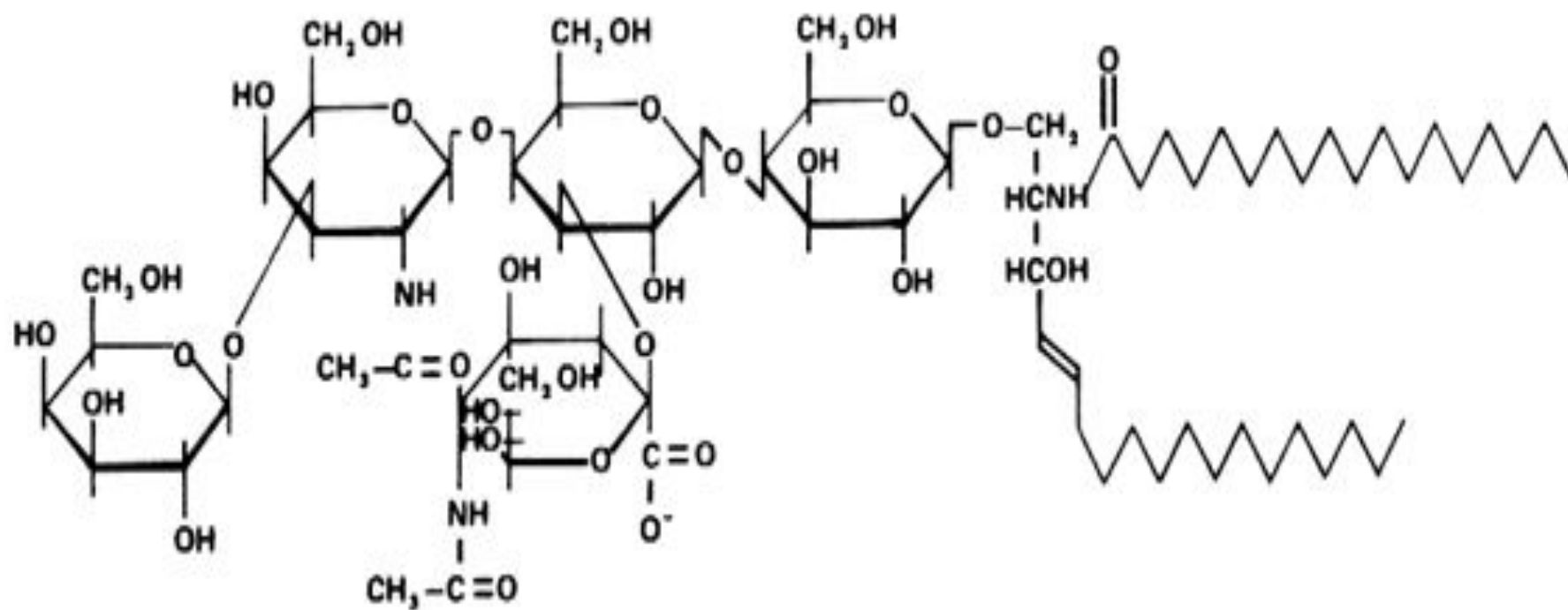


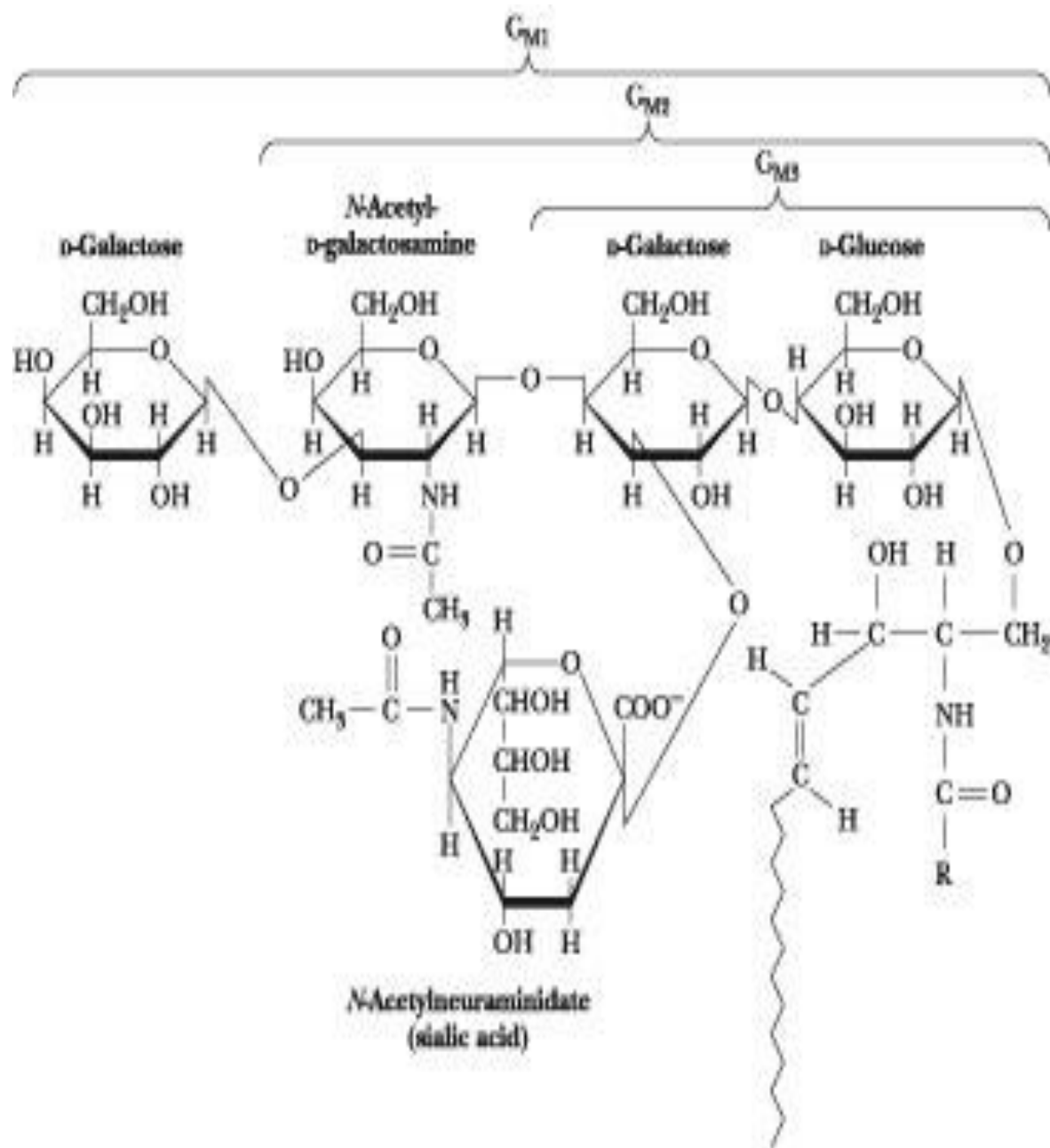
3:GANGGLIOSIDES:



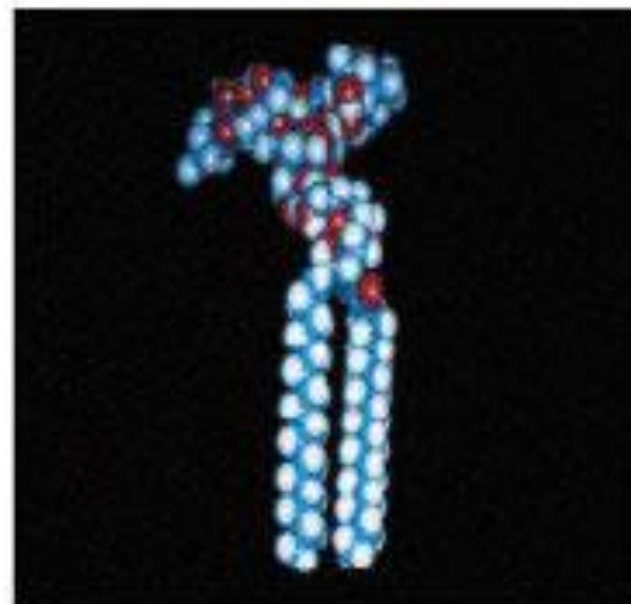
-

Gangliosides are the group of sphingolipids that show the greatest structural variation and also the more complex structure. Gangliosides are characteristic of vertebrate nervous tissues. This group includes molecules composed of ceramide linked by a glycosidic bond to an oligosaccharide chain containing hexose and **N-acetylneuraminic acid** (NANA, acidic sugar known also as **sialic acid**) units.

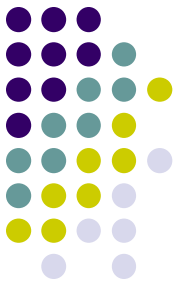




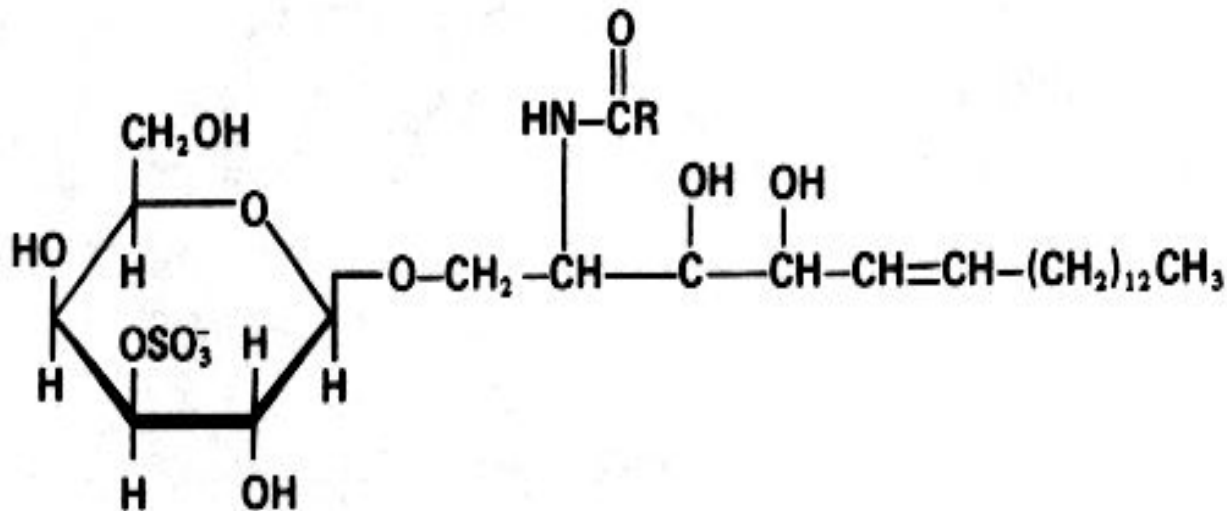
Gangliosides G_{M1} , G_{M2} , and G_{M3}

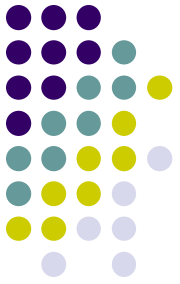


4:SULFOLIPIDS OR SULFATIDES



- **Sulfolipids** are a class of lipids which possess a sulfur-containing functional group .
- This group (known also as **cerebroside 3-sulfate**) is mainly formed of 3-sulfate esters of galactosylcerebrosides (galactosyl-3-sulfate esters) .





THANK YOU